Shaik Sattar Saif

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	9.09
Narayana Junior College	Class XII (Telangana State Board of Intermediate)	2023	98.40%
Narayana School	Class X (Board of Secondary Education, Telangana)	2021	10.00%

WORK EXPERIENCE

Oracle | *Software Developer Intern*

- **Formulated drone dispatch as a spatial temporal constrained decision problem**, jointly evaluating route feasibility, time varying environmental conditions, battery availability, payload constraints, and delivery deadlines.
- **Implemented time dependent Dijkstra and Yen's K-shortest-path algorithms** over dynamically generated geographic graphs, with edge costs and route validity varying across temporal weather and airspace conditions.
- **Developed an iterative constraint propagation procedure** to determine feasible departure times under coupled route, battery recharge, weather clearance, and SLA constraints, with staged allocation to avoid premature dispatch.
- **Built a discrete event simulation engine** with virtual time execution and deterministic scenario control for evaluating asynchronous arrivals, resource contention, scheduling decisions, and drone/battery state transitions.
- **Integrated external weather and geospatial data using lazy refresh caching and spatial indexing**, reducing redundant data retrieval while providing environmental inputs for route and dispatch decisions.

PROJECTS

Early Prediction of Ventilation Failure in ICU Patients with COPD | *Research Project under Dr. Rahuldeb*

- Architected a **DAG-based CXR feature extraction framework** on **MIMIC-CXR** with persistent caching, deriving 25+ quantitative radiological biomarkers from anatomical segmentation.
- Adapted **HybridGNet via transfer learning** for diaphragm landmark regression, redesigning its topology as a **20-node open-graph Chebyshev GCN** with a composite geometric loss.
- Contributed CXR-derived features to the project's multimodal model, improving ventilation-failure **AUROC from 0.8482 to 0.9407** and **AUPRC from 0.5177 to 0.7570.**

Crypto Trade Analysis | GitHub link

- Built a **Hierarchical Negative Binomial regression model** to capture severe trade count overdispersion and hourly patterns, reducing 95% predictive interval outlier rates from **85.5% (Gamma Poisson baseline) to 4.13%**.
- Implemented a **custom component wise Metropolis within Gibbs MCMC sampler from scratch** for posterior inference over 32 parameters, including 24 hourly hierarchical intercepts and market microstructure covariates.
- Modeled trade volume using a **conjugate Normal Inverse Gamma framework**, achieving **5.15%** outlier rates on nominal 95% predictive intervals and deriving closed-form **Bayes Factors (BF)** to test whether mean volume remained within a defined equilibrium range across varying lookback horizons.

Optimal Redundancy Allocation in Dependent Series-Parallel Systems | *Research with Prof. Sameen Naqvi*

- Developed a **distortion-function framework** for optimizing redundancy allocation in two-component series and parallel systems with **dependent component lifetimes**.
- Derived sufficient conditions to compare allocation strategies under **usual stochastic, hazard-rate, and reversed-hazard-rate orders**, incorporating redundancy mechanisms, lifetime distributions, and copula-based dependence.
- Validated the theoretical results using **Weibull, Pareto, and Gamma models** and a real-data application to **high-voltage transmission cable reliability**.

Lumin - Scientific Document Research Assistant

- Built the initial **scientific document RAG system from scratch**, implementing custom FAISS vector indexing, **section aware token chunking**, metadata filtering, reranking, and retrieval to generation pipelines.
- Evolved the system into a structured document intelligence platform with **geometric layout analysis**, **multi column reading order reconstruction**, and a **provenance aware document AST** linking content to PDF coordinates.
- Engineered a **modular QA and generation layer** with multi-strategy prompting, citation validation, **confidence scoring**, and a vendor-agnostic interface across multiple LLM providers.

SKILLS

- **Technical Skills**: Python, C++, SQL, PyTorch, Scikit-learn, NumPy, Pandas, SciPy, Git.
- **ML & Statistics**: CV, Image Segmentation, Transfer Learning, Embeddings, Statistical Modeling, Bayesian Inference.

RELEVANT COURSES

- **Mathematics**: Probability, Linear Algebra, Applied Statistics, Bayesian Statistics, Statistical Inference, Stochastic Processes, Graph Theory, Spectral Graph Theory.
- **Computer Science**: Data Structures and Applications, Algorithms, DBMS.
- **Artificial Intelligence**: Pattern Recognition and Machine Learning, Deep Learning (from Coursera), Natural Language Processing, Reinforcement Learning.

EXTRACURRICULAR

- **Core Member of Epoch** - AIML club of IITH
- **Class Representative**, 2024-2025
- **Volunteer**, NSS IITH- 141+ hours of community service